

Exercise1 Control Structures

-- Create Database

```
CREATE DATABASE bank_db;  
USE bank_db;
```

-- Create Customers Table

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100),  
    age INT,  
    balance DECIMAL(10,2),  
    is_vip CHAR(1) DEFAULT 'N'  
);
```

-- Create Loans Table

```
CREATE TABLE loans (  
    loan_id INT PRIMARY KEY,  
    customer_id INT,  
    interest_rate DECIMAL(5,2),  
    due_date DATE,  
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
);
```

-- Insert Sample Customers

```
INSERT INTO customers VALUES  
(1, 'Ramesh Kulkarni', 65, 12000, 'N'),  
(2, 'Sneha Patil', 45, 8000, 'N'),  
(3, 'Arvind Joshi', 70, 15000, 'N'),  
(4, 'Priya Deshmukh', 32, 5000, 'N');
```

-- Insert Sample Loans

```
INSERT INTO loans VALUES  
(101, 1, 9.50, CURDATE() + INTERVAL 10 DAY),  
(102, 2, 8.00, CURDATE() + INTERVAL 45 DAY),  
(103, 3, 10.20, CURDATE() + INTERVAL 5 DAY),  
(104, 4, 7.50, CURDATE() + INTERVAL 60 DAY);
```

-- View Initial Data

```
SELECT * FROM customers;
```

```
SELECT * FROM loans;
```

```
-- =====
```

```
-- Scenario 1:
```

```
-- Discount loan interest rate by 1%
```

```
-- for customers older than 60
```

```
-- =====
```

```
UPDATE loans l
```

```
JOIN customers c
```

```
ON l.customer_id = c.customer_id
```

```
SET l.interest_rate = l.interest_rate * 0.99
```

```
WHERE c.age > 60;
```

```
-- Check Result
```

```
SELECT * FROM loans;
```

```
-- =====
```

```
-- Scenario 2:
```

```
-- Mark customers as VIP if balance > 10000
```

```
-- =====
```

```
UPDATE customers
```

```
SET is_vip = 'Y'
```

```
WHERE balance > 10000;
```

```
-- Check Result
```

```
SELECT * FROM customers;
```

```
-- =====
```

```
-- Scenario 3:
```

```
-- Loan reminders due within 30 days
```

```
-- =====
```

```
SELECT
```

```
  CONCAT(
```

```
    'Reminder: ',
```

```
    c.customer_name,
```

```
    ' has loan ',
```

```
    l.loan_id,
```

```
        ' due on ',  
        DATE_FORMAT(l.due_date,'%d-%b-%Y')  
    ) AS Reminder_Message  
FROM loans l  
JOIN customers c  
ON l.customer_id = c.customer_id  
WHERE l.due_date BETWEEN CURDATE()  
AND CURDATE() + INTERVAL 30 DAY;
```

```
-- Final Data  
SELECT * FROM customers;  
SELECT * FROM loans;
```